Limit Cycles of Model-Free PID Tuning by Step-Response Inspection

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***Abstract*— SPIN tunes a PID controller from closed-loop step responses: it counts how often three phase portraits of the error turn around the origin and moves the gains by a fixed table of four moves. In every run we made, the iteration ends in a periodic orbit. We characterize such orbits in the deterministic case. Every closed orbit that stays below the upper gain bound, whatever turn index drives the rule, applies the four moves in the proportions $1:4:2:1$, has a period of $8n_e$ for an integer multiplicity n_e, and stays within $n_e(2,1,\frac{1}{2})$ steps of its centre in $(\log K_i, \log K_p, \log K_d)$. Its bounding box meets the switching boundaries, so as the step shrinks the orbits accumulate where the boundaries meet. For the published turn count this set is not a point. The count is the limit of a continuous count that makes the rule's move on 98.5–$99.5\,\%$ of the states the rule visits, and through it the destination becomes visible: all 108 terminal orbits on a four-plant battery straddle the event at which the overshoot of the step response equals the settling band $\delta = 2\,\%$, and every quiet step has a smaller overshoot. This is a damping condition on the step response of the sensitivity function S. The orbits therefore differ in the gains but hardly in robustness: the largest sensitivity peak on an orbit lies within 1.54–2.10 when the derivative acts on the error, and reaches 2.95 when it acts on the measurement and the record is no longer the step response of S. A plant-free step-size rule and a drift meter follow from the orbit structure.**

I. Introduction

Most PID tuning methods first fit a model and then compute the gains [3], [4]. SPIN [1] tunes from what the closed-loop step response looks like instead. It draws three phase portraits of the control error, counts how often each turns around the origin, compares the three *turn counts* N_k with fixed thresholds $\bar{N}_k$, and applies one of four moves to the gains. No model is identified. Iterative feedback tuning [5] and extremum seeking [6] estimate gradients of a cost; the triangular rule uses only three signs. Its natural end state is therefore not a stationary point but a periodic orbit of a switched system, and this paper is about that orbit.

We ask where the orbit is, what shape it has, what happens to it as the step shrinks, and why it ends at a robust controller. The answers come in three layers. The first is combinatorial and holds for any turn index: the move counts, the period and the excursion of every closed orbit (Section III). The second is topological and also holds for any index: the orbit's box meets the switching boundaries, so small-step orbits accumulate on their intersection (Section IV). The third concerns the published count, a winding number over a window that ends when the error settles, which jumps when that window jumps. We construct a continuous count of which it is the limit, find that the orbits sit on the event at which the overshoot equals the settling band δ, and read that event as a damping condition on the sensitivity function (Sections V–VI). This explains why the destination is not unique and why it is robust anyway. Section VII turns the orbit structure into a step-size rule and a stopping test.

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II. The Rule as a Switched System

Loop and record. A discrete-time loop with sampling period T_s closes a parallel PID controller $K_p + K_i/s + K_d s/(1 + (K_d/\nu)s)$, $\nu = 10$, around the plant, and a unit setpoint step is applied at $t = 0$. The record is the error $e_n = r - y_n$, $n = 0, \dots, n_{\max}$.

Assumption 1: The loop is linear and noise-free, and the derivative acts on the error.

Under Assumption 1 the increments $e_n - e_{n-1}$ ($e_{-1} = 0$) are the impulse response of the sensitivity function S, so the record is the step response of S [2]. We return to this in Section VI.

Turn counts. From the record, SPIN draws three portraits: $\Gamma_0 = (e, E)$ with E the running sum of e, $\Gamma_1 = (\Delta e, e)$ and $\Gamma_2 = (\Delta^2 e, \Delta e)$. They are read over a window $[k_1, k_\delta]$. The start k_1 is the first sample at which the output has risen past $10\,\%$ of the step, plus a fixed padding. The end k_δ is one sample past the last sample with $|e_n| > \delta \max_m |e_m|$, $\delta = 0.02$, clipped to $[k_1, k_2]$, where k_2 is the end of the record less a padding. Each axis is normalized by its peak over the window, the integral axis of Γ_0 is taken relative to its value at k_δ, and the turn count N_k is the winding number of Γ_k around the origin, up to the last segment that touches the disc of radius $\varepsilon = 0.1$. The thresholds are $\bar{N} = (0.5, 0.75, 1.0)$.

Moves. We write $\theta = (\log K_i, \log K_p, \log K_d)$, in band order. If the stability screen of [1] finds the record growing, all gains are backed off by $b = (1, 2, 3)\log 2$. Otherwise the lowest band with $N_k > \bar{N}_k$ is cut and the bands below it are raised; if no band is violated, all are raised. With $h = \log\gamma$, $\gamma = 1/(1-\beta)$, the four moves are translations by h times

$$\begin{aligned} g &= (1,1,1), & d_0 &= (-1,0,0), \\ d_1 &= (1,-1,0), & d_2 &= (1,1,-1). \end{aligned} \tag{1}$$

A g-step ("grow") is *quiet*: it is taken only where no band is violated. The gains live in the box $\Theta = [\ell, u]^3$, $\ell = \log 10^{-3}$, $u = \log 10$; raises are clipped at u and cuts at ℓ. We use $\beta = 0.1$ unless stated.

Two corrections to [1]. (i) A cut is *blocked* if its gain is already at ℓ. A blocked cut now passes to the nearest lower

band whose gain can still be cut, with the raises below that band; if there is none, the gains are held. Previously the raises were applied although the cut was not, which applied the vector $(1, 1, 0)$, outside the table (1). (ii) The winding number is computed as the sum of the signed angles of the portrait's segments. The published implementation corrected the endpoint angle by crossings of the negative real axis but missed crossings by a segment with one end at $x \geq 0$, and on the last segment; it miscounted by one or two full turns on 41 of the 8792 states visited in Section V. Both corrections are in the reference code.

Experiments. The battery is that of [1]: P1 $e^{-s}/((10s+1)(s+1)^3)$, P2 $1.25e^{-8s}/(5s+1)^4$, P3 $e^{-10s}/((2s+1)(s+1)^3)$, P4 $e^{-4s}/(8s+1)^6$. Each record has 500 samples over ten times the sum of the plant's time constants and dead time. For each plant we start from the AMIGO gains [4] and from the 26 gain vectors obtained by multiplying each AMIGO gain by 8, 1 or 1/8, and run the deterministic iteration until it revisits a state. The states from the first visit on form the terminal orbit.

III. Closed Orbits

Lemma 1: Let $\log 2/h$ be irrational. On a closed orbit of period $T \geq 2$ that does not reach the ceiling u and on which K_i stays above ℓ, every step is a full table move $\theta^{t+1} = \theta^t + hv^t$, $v^t \in \{g, d_0, d_1, d_2\}$.

Proof: A step is a table move, possibly with a cut clipped at ℓ; a raise clipped at u; a screen backoff, possibly clipped at ℓ; or a hold. A hold is a fixed point ($T = 1$), and clipped raises are excluded. The coordinate $\log K_i$ is never clipped, so its steps are 0, $\pm h$ or $-\log 2$; closing the orbit gives $mh = n \log 2$ with n the number of screen events, hence $n = 0$. In $\log K_p$ and $\log K_d$ the remaining steps are 0, $\pm h$ and cuts clipped to ℓ. If a coordinate takes the value ℓ on the orbit, then following the orbit from there it takes only values in $\ell + h\mathbb{Z}$, since $\pm h$ preserves this set and a clipped cut would start in $(\ell, \ell + h)$; so no cut is clipped. If it never takes the value ℓ, no cut is clipped either. By correction (i), a passed-on cut is a table move of a lower band. □

For $\beta = 0.1$, $\log 2/\log(10/9)$ is irrational: $2^q = (10/9)^p$ would give $2^q 3^{2p} = 2^p 5^p$.

Proposition 1 (Balance): The only convex combination of the moves (1) that vanishes is $\frac{1}{8}g + \frac{1}{2}d_0 + \frac{1}{4}d_1 + \frac{1}{8}d_2 = 0$.

Proof: With weights w, the $\log K_d$ coordinate gives $w_g = w_2$, then $\log K_p$ gives $w_1 = 2w_g$, $\log K_i$ gives $w_0 = 4w_g$, and $\sum w = 1$ gives $w_g = \frac{1}{8}$. □

Theorem 1 (Closed orbits): Under the hypotheses of Lemma 1, the orbit applies the moves (g, d_0, d_1, d_2) exactly $n_e(1, 4, 2, 1)$ times for an integer $n_e \geq 1$, its *multiplicity*; so $T = 8n_e$ and every move occurs. Each coordinate stays within $n_e(2, 1, \frac{1}{2})\, h$ of the midpoint of its range on the orbit; we call the vector of these midpoints the orbit's *centre*.

Proof: By Lemma 1 the counts n_v satisfy $\sum_v n_v v = 0$; the computation of Proposition 1 over the integers gives $n = n_e(1, 4, 2, 1)$, and $T \geq 2$ forces $n_e \geq 1$. A closed walk with m up-steps and m down-steps of size h has range at most mh; per period, $\log K_i$, $\log K_p$ and $\log K_d$ are raised $4n_e$, $2n_e$ and n_e times. □

On the battery, the six P4 orbits that reach the K_d ceiling are excluded by the hypotheses; five of them have period four. All other 102 orbits have the predicted counts and lie within the bound; 93 have $n_e = 1$, the others $n_e \in \{2, 3, 4, 5, 9, 24\}$. Three of P1's, thirteen of P2's and fifteen of P4's orbits have K_d on the floor, where passed-on cuts keep the table intact. From the AMIGO start every plant closes an orbit of period eight after 9 to 19 experiments.

The table (1) is not an arbitrary choice.

Proposition 2 (The table is forced): Among tables for n bands with entries in $\{-1, 0, 1\}$, $g = (1, \ldots, 1)$ and $d_k[k] = -1$, the staircase table ($d_k[j] = 1$ for $j < k$, 0 for $j > k$), which is (1) for $n = 3$, is the only one whose balance weights equal the frequencies with which the priority logic selects the moves when the n signs $N_k \gtrless \bar{N}_k$ are equally likely.

Proof: The logic selects g on 1 of the 2^n sign patterns and d_k on 2^{n-k-1}. Matching requires $g + \sum_k 2^{n-k-1} d_k = 0$; coordinate j reads $\sum_k 2^{n-k-1} d_k[j] = -1$, a signed-binary representation of -1. The n such representations have a single -1 digit at a position m, with $d_k[j] = +1$ for $k > m$ and 0 for $k < m$. The hypothesis $d_j[j] = -1$ forces $m = j$, which is the staircase. □

A consequence for noise: if noise makes the comparisons independent fair coins, the moves occur at the balance weights and the expected displacement is zero. The balance point does not drift; only the spread grows.

IV. Localization and the Small-Step Limit

Let Σ_k be the boundary of the set $\{\theta \in \Theta : N_k(\theta) > \bar{N}_k\}$: the switching boundary of band k. No continuity of N is assumed.

Proposition 3 (Localization): Let a closed orbit satisfy Theorem 1, and let B be its bounding box. Then B meets Σ_2. If no cut on the orbit is passed on, B meets Σ_0, Σ_1 and Σ_2.

Proof: The orbit contains a quiet step, taken where $N \leq \bar{N}$, and a d_2-step, taken only where $N_2 > \bar{N}_2$. Without passed-on cuts, a d_k-step is taken only where $N_k > \bar{N}_k$. The box is connected and contains points of $\{N_k > \bar{N}_k\}$ and of its complement, so it meets the boundary. □

Corollary 1 (Small-step limit): Let $\beta_j \to 0$, and let the orbits O_j satisfy Proposition 3 without passed-on cuts, with $n_e^{(j)} h_j \to 0$. Every accumulation point of O_j lies in $\mathcal{T} = \Sigma_0 \cap \Sigma_1 \cap \Sigma_2$.

Proof: The diameter of B_j is at most $\sqrt{21}\, n_e^{(j)} h_j \to 0$ by Theorem 1; each B_j meets each closed set Σ_k. □

Remark 1 (Transversal case): If N is C^1 near a point θ^* with $N(\theta^*) = \bar{N}$ and a nonsingular Jacobian J^*, then $\mathcal{T}$ is locally $\{\theta^*\}$. A box B near θ^* that meets the three level sets satisfies $|N_k - \bar{N}_k| \leq \|\nabla N_k\| \operatorname{diam} B$ at its centre by the mean-value theorem, and the inverse function theorem turns this into a distance $O(\|(J^*)^{-1}\| \|J^*\| \operatorname{diam} B)$ from θ^*. For orbits of bounded multiplicity without passed-on cuts this is a rate $O(h)$.

The existence and attraction of a terminal orbit remain open; what the results above fix is where any closed orbit can be, and where small-step orbits must end up. On the battery, from the AMIGO start, halving β from 0.1 to 0.00625 moves the orbit centre by 0.39, 0.031, 0.015 and 0.007 on P1 and by 0.18, 0.041, 0.020 and 0.002 on P2 (in log-gain): the centres converge. On P3 the centre jumps once (0.25 at $\beta = 0.025$) and then moves by less than 0.01; on P4 the orbits reach the K_d ceiling for $\beta \leq 0.05$. What $\mathcal{T}$ looks like depends on the count.

V. The Published Count as a Limit

The published count is a composition of discrete steps: the window start k_1, the window end k_δ, the peak normalization over the window, the reference value of Γ_0's integral axis at k_δ, and the cut at the last disc contact. Each makes N jump. We replace each by a smooth version. Let ψ be the logistic function and $A_n = \max_{m \geq n} |e_m| / \max_m |e_m|$ the *future envelope* of the error. Sample n is before the window end with weight $P_n = \psi(p \log(A_{n-1}/\delta))$, the smooth version of $n \leq k_\delta$, and after the window start with weight $Q_n = \psi(p_s(0.9 - \min_{m \leq n-\kappa} e_m))$, κ the padding. Each portrait axis is normalized by $\max_n Q_n P_n |a_n|$, and Γ_0's reference value is $\sum_n (P_n - P_{n+1}) E_n$. Segment n of a portrait, from (a_n, b_n) to (a_{n+1}, b_{n+1}), turns by $\varphi_n = \operatorname{atan2}(a_n b_{n+1} - b_n a_{n+1},\ a_n a_{n+1} + b_n b_{n+1})$ and counts with the weight $\omega_n = Q_n P_{n+1} V_n \rho_n^2 / (\rho_n^2 + r_0^2)$. Here ρ_n is the segment's distance from the origin, and V_n is the smooth version of "a later segment touches the disc, or no segment does", built from $\psi(p \log(\varepsilon/\rho_m))$. The continuous count of band k is

$$N_{k,p} = \frac{1}{2\pi} \sum_n \omega_n \varphi_n \tag{2}$$

over the portrait Γ_k. We use $p = 32$, $p_s = 200$ and $r_0 = \varepsilon/100$.

Proposition 4 (Continuity and limit): For finite p, p_s and $r_0 > 0$, N_p is a continuous function of the record wherever the record and the portrait axes are not identically zero; under Assumption 1 it is therefore continuous in θ. As $p, p_s \to \infty$ and $r_0 \to 0$, N_p tends to the published count at every record without ties ($|e_n| = \delta \max |e|$, the output exactly at the 10 % level, a segment at distance exactly ε or through the origin), except when the last disc contact is on the second segment, where the published count is set to zero.

Proof: Envelopes, extrema, logistics and products are continuous. The segment angle is discontinuous only where the segment passes through the origin, where $\rho_n = 0$ and its weight vanishes, so the product is continuous. A finite linear recursion makes the record continuous in the gains. As the sharpnesses grow, each logistic tends to the indicator it smooths, off the ties. □

Table I measures the limit and the agreement on the states visited in the first 120 iterations from each of the 27 starts. The hard limit coincides with the published count on 8791 of 8792 states, the exception being the second-segment case. At $p = 32$ the continuous count makes the rule's move on 98.5–99.5 % of the states, and driving the rule with it reaches the same terminal orbit from 67–100 % of the starts; from the others it reaches another orbit of the kind described next.

TABLE I

The continuous count against the published one, on the states visited in the first 120 iterations from each of 27 starts per plant.

	states	hard limit exact	same move $p = 8$	same move $p = 32$	same orbit $p = 32$
P1	1926	1926	95.6 %	98.8 %	20/27
P2	2332	2332	96.2 %	98.9 %	21/27
P3	2216	2215	96.4 %	98.5 %	18/27
P4	2318	2318	97.6 %	99.5 %	27/27

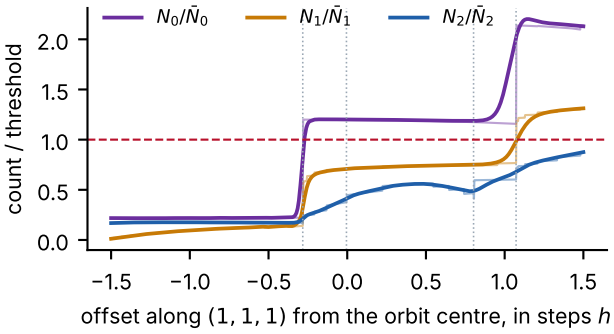


Fig. 1. The three counts along the direction $(1, 1, 1)$ through the centre of P1's terminal orbit from the AMIGO start, published (thin) and continuous (thick, $p = 32$), divided by their thresholds. At $-0.28h$ the overshoot reaches δ, the window end k_δ jumps past the overshoot lobe, and all three counts jump with it; N_0 crosses its threshold there. Dotted: jumps of the published count.

Where the orbits sit. Let the overshoot be the largest excursion of the output above the setpoint, relative to $\max |e|$. While it is below δ, the window ends before the overshoot lobe and the portraits do not contain it; once it exceeds δ, the window end k_δ jumps past the lobe, and since the window is shared by the three portraits, all three counts jump together (Fig. 1). This *overshoot event* is where every orbit sits. All 108 terminal orbits straddle it: each has points with overshoot below and above δ. Every quiet step on every orbit has an overshoot below δ (at most 1.96 %), while 68–75 % of the corrective steps, by plant, have one above it (median 3.5–7.6 %). Where the jump carries a count across its threshold, as for N_0 in Fig. 1, the switching boundary of that band is a piece of the event surface.

Consequence: the destination is not a point. With jumps, $\mathcal{T}$ is not the zero set of a continuous map, and Remark 1 does not apply to the published count. Moreover the counts determine K_d only weakly. The distinct terminal orbits of a plant spread over 5.0–9.2 in $\log K_d$ but only 0.25–0.69 in $\log K_i$ and $\log K_p$, and the continuous count's Jacobian at the orbit centres away from the K_d bounds has a median $\sigma_{\min}/\sigma_{\max}$ of 0.009–0.034 by plant (on P4 a single such centre). The start selects the orbit: 4 to 9 distinct orbits per

plant (Table II).

VI. Robustness

The orbits differ, but not in robustness (Table II): over all 108 runs the largest sensitivity peak M_s on an orbit lies within 1.54–2.10. Among 500 states sampled from the first 60 iterations of the runs, M_s does not exceed 1.64–1.88 (by plant) on those where all three counts are within their thresholds, and the rank correlation of $\max_k N_k/\bar{N}_k$ with M_s over all stable sampled states is 0.70–0.84. Three observations explain this.

The event is a damping condition on S. Under Assumption 1 the record is the step response of S, so the counts and the window are functionals of S alone. The orbits sit where the overshoot of that step response equals δ. For a dominant pole pair with damping ζ the overshoot is $\exp(-\pi\zeta/\sqrt{1-\zeta^2})$, so the event is a damping level, $\zeta \approx 0.78$ for $\delta = 0.02$, and a damping level is a robustness specification. In this setting the quiet region lies inside $\{\text{overshoot} < \delta\}$: the settling band δ, more than the thresholds $\bar{N}$, fixes how damped the destination is. The continuous count of Section V is this condition made smooth: its window weight P_n is a logistic in $\log(A_{n-1}/\delta)$.

The record must be the step response of S. If the derivative acts on the measurement, the setpoint step reaches the plant through the proportional and integral terms only, and the record is the step response of $(1 + PC_D)S$, with C_D the derivative term, not of S. Its overshoot is then no damping condition on S. On the same starts, M_s reaches 2.95 on P1, 2.41 on P2 and 2.35 on P3 (Table II). No safeguard in the rule is needed when the record is the step response of S. If the derivative must stay on the measurement, a step ξ added to the measured signal gives $e = -S\xi$, because the controller then sees the whole step. Assumption 1 is also what the certified bound on M_s from the same record requires [2].

Robustness of the decisions. Near the event the published count jumps when the overshoot crosses δ, so a perturbation of the record of size $\eta \max|e|$ can flip a decision when the overshoot lies within η of δ. The continuous count replaces the jump ΔN by a slope of order $p\,|\Delta N|/4$ in $\log(A/\delta)$: p sets how sharply the decisions respond to perturbations of the record, and δ sets their scale. A small p trades fidelity to the published rule (Table I) for decisions that change continuously with the record.

VII. Choosing the Step and Stopping

Corollary 2 (Step rule): On an orbit with $n_e = 1$, K_i, K_p and K_d stay within the factors γ^2, γ and $\gamma^{1/2}$ of the orbit centre. To hold every gain within a factor ρ it suffices that $\beta \le 1 - \rho^{-1/2}$.

The rule contains no plant parameter. At $\beta = 0.1$ the factors are 23.5 %, 11.1 % and 5.4 %. The centre is one of several, all with similar robustness; the rule bounds the excursion around the one the start selects.

Proposition 5 (Drift meter): Over a window of iterations that apply table moves at a fixed step h, let s be the share of each move and M the matrix with rows (1). The mean displacement per iteration is $\bar{v} = hM^\top s = hM^\top(s - w)$, $w = (\frac{1}{8}, \frac{1}{2}, \frac{1}{4}, \frac{1}{8})$, and $\sqrt{2}\,h\|s-w\| \le \|\bar{v}\| \le 1.841\,h\|s-w\|$.
Proof: $M^\top w = 0$ by Proposition 1. Since $s - w$ sums to zero, the constants are the extreme singular values of $M^\top$ on that subspace, $\sqrt{2}$ and 1.841. □

TABLE II

Terminal orbits from 27 starts per plant: distinct orbits, range of K_i (mean over each orbit), range over the orbits of the largest M_s on each, orbits with K_d on the floor/ceiling, and the largest M_s with the derivative on the measurement, same starts.

	orbits	K_i	M_s	K_d	M_s, D on y
P1	9	0.122–0.220	1.54–1.79	3/0	2.95
P2	7	0.024–0.038	1.73–1.97	13/0	2.41
P3	6	0.054–0.070	1.88–2.10	0/0	2.35
P4	4	0.016–0.028	1.70–1.84	15/6	1.81

The meter reads the progress of the rule from the moves it has made, without an experiment. By Theorem 1 it reads zero over every full period of a closed orbit, which is a stopping test: stop once it reads zero over a period at the step one is willing to keep. While the gains travel it does not read zero, and the step should not shrink.

Lemma 2 (Logarithmic travel): With $\beta_n = c_0/n$, $0 < c_0 < 1$, the distance covered in n steps is at most $-\sum_{m\le n}\log(1 - c_0/m) = c_0 \log n + O(1)$, so crossing a distance D takes at least $e^{D/c_0 - O(1)}$ steps.

A schedule should therefore shrink the step only when the gains have stopped travelling. We halve h once every band has moved both up and down since the last halving. From the AMIGO start the step falls below 10^{-3} of its initial value after 153 (P1), 153 (P2) and 224 (P3) experiments, at M_s of 1.70, 1.76 and 1.77; on P4 the run reaches the K_d ceiling. Between two halvings K_d has moved up and down, so a quiet step and a d_2-step were taken and, as in Proposition 3, the box of those iterates meets Σ_2. If the halvings recur indefinitely and the iterates converge, the limit therefore lies on Σ_2, and on all three boundaries if no cut is passed on. In our runs the halvings recur on P1 and P3 (34 and 19 in 600 experiments); on P2 the last halving is at experiment 496 and the run ends in a stretch of d_1 moves, during which K_d does not move and the step cannot shrink.

Remark 2 (Travel certificate): By Theorem 1, a run of more than $n_e\kappa_v$ consecutive moves v, $\kappa = (1, 4, 2, 1)$, cannot occur on a closed orbit of multiplicity n_e or less. Such a run certifies travel, and the step may be enlarged during it without altering any such orbit. Doubling on every repeated move alters every orbit whose ordering repeats a move.

VIII. Conclusions

The limit cycle of SPIN has a combinatorial core that does not depend on how the turns are counted: the move shares $(\frac{1}{8}, \frac{1}{2}, \frac{1}{4}, \frac{1}{8})$, periods $8n_e$, an excursion of $n_e(2, 1, \frac{1}{2})$ steps, and a bounding box that meets the switching boundaries, so that small-step orbits end on their intersection. This needs

every applied move to belong to the table, which is why a blocked cut must pass to a lower band rather than raise the bands below it.

Where the boundaries meet depends on the count. The published count is a winding number over a window that ends when the error settles, and the limit of a continuous count that reproduces the rule's decisions on 98.5–$99.5\,\%$ of the states it visits. Through that count the destination becomes visible: every terminal orbit on the battery sits on the event at which the overshoot of the step response equals the settling band. The destination is therefore not a point, and K_d in particular is weakly determined; but the event is a damping condition on S, and the orbits share its robustness, M_s within 1.54–2.10. That reading holds only when the record is the step response of S, the same condition under which the record certifies M_s [2]. The existence and attraction of the terminal orbit, and the stochastic case, remain open.

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